8.FOURIER TRANSFORM AND FREQUENCY DOMAIN ANALYSIS

Experiment 8.1

Fourier Transform of Periodic Impulse Train .

Aim:

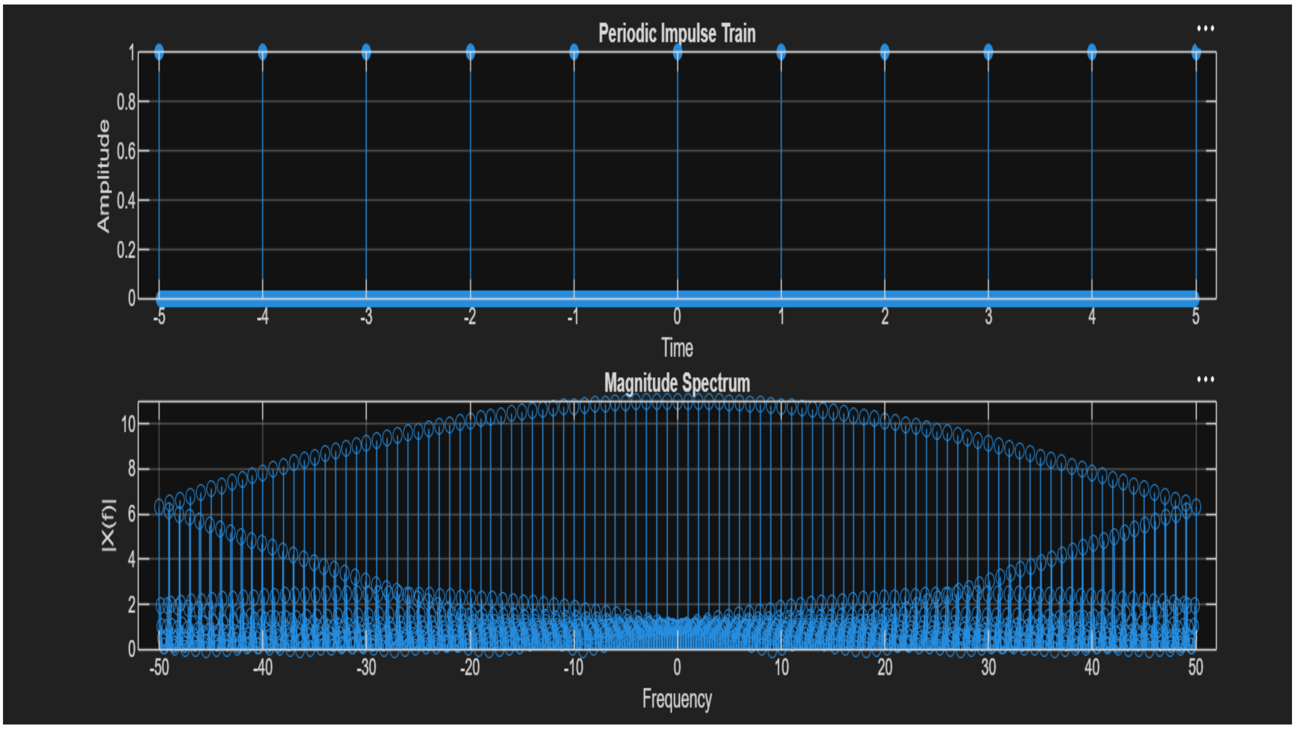
To compute and analyze the Fourier Transform of a periodic impulse train using MATLAB. The Aim is to observe the frequency-domain characteristics of a periodic impulse train, which should result in a discrete set of frequency components due to the periodicity of the signal.

Program:

```
% Fourier Transform of Periodic Impulse Train

clc;

clear;

close all;

% Time axis

t = -5:0.01:5;

% Generate periodic impulse train

x = zeros(size(t));

x(1:100:end) = 1; % Impulse every 100 samples

% Fourier Transform

X = fftshift(fft(x));

% Frequency axis

N = length(X);

f = linspace(-50, 50, N);

% Plotting

figure;
```

```
subplot(2,1,1);

stem(t, x, 'filled');

grid on;

title('Periodic Impulse Train');

xlabel('Time');

ylabel('Amplitude');

subplot(2,1,2);

stem(f, abs(X));

grid on;

title('Magnitude Spectrum');

xlabel('Frequency');

ylabel('|X(f)|');
```

Output waveform:



Result:

The Fourier Transform of a periodic impulse train was obtained using MATLAB. The frequency-domain representation was observed to be another impulse train with discrete spectral lines, confirming the Fourier Transform property of periodic impulse trains and illustrating the relationship between time-domain periodicity and frequency-domain discreteness.

Experiment 8.2

Discrete Time Fourier Transform.

Aim:
The Aim of this experiment is to understand and implement the Discrete Time Fourier Transform (DTFT) using MATLAB. The DTFT is a fundamental concept in signal processing, used to analyze the frequency content of discrete-time signals. This experiment will involve:

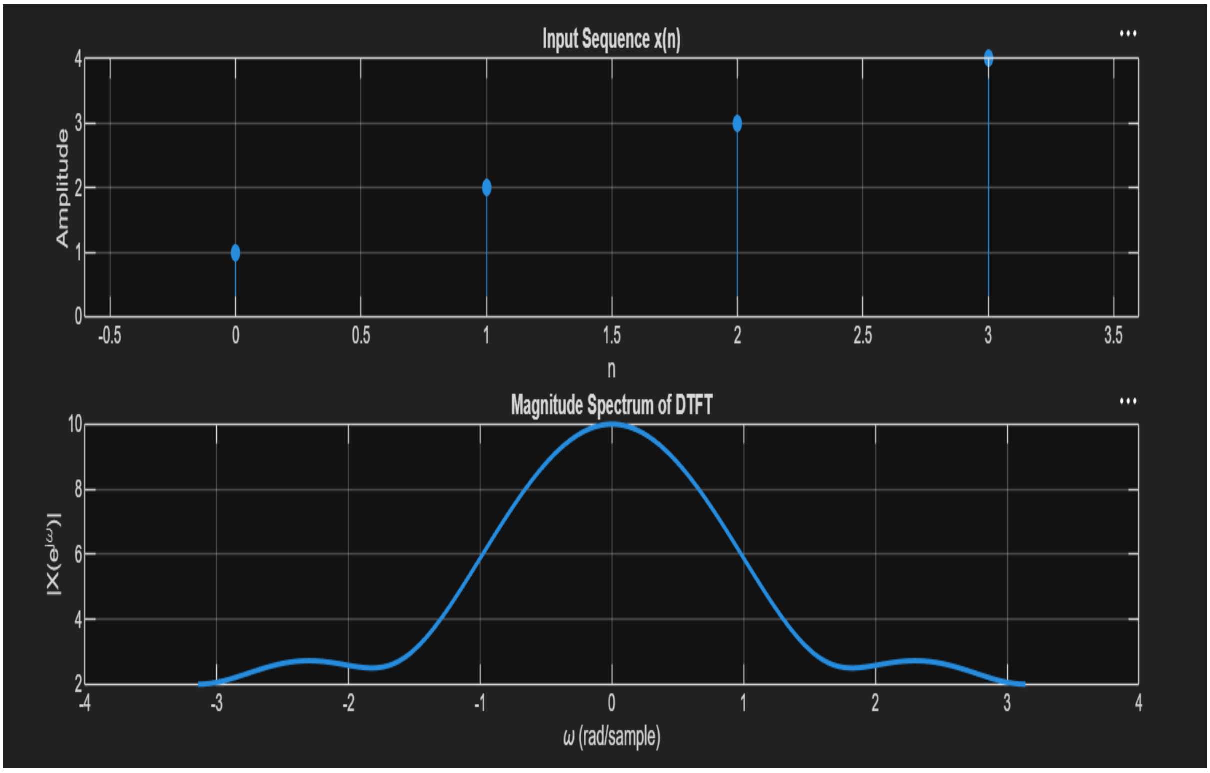
1. Computing the DTFT of a given discrete-time signal.
2. Plotting the magnitude and phase response of the DTFT.
3. Analyzing the effects of signal length and frequency components on the DTFT.

Program:

```
% Discrete-Time Fourier Transform (DTFT)

clc;

clear;

close all;

% Input sequence

x = [1 2 3 4];

% Frequency range

w = -pi:0.01:pi;

% DTFT computation

X = zeros(size(w));
```

```
for k = 1:length(w)

    for n = 1:length(x)

        X(k) = X(k) + x(n) * exp(-1j * w(k) * (n-1));

    end

end


% Plotting

figure;

subplot(2,1,1);

stem(0:length(x)-1, x, 'filled');


grid on;

title('Input Sequence x(n)');

xlabel('n');

ylabel('Amplitude');

subplot(2,1,2);


plot(w, abs(X), 'LineWidth', 2);

grid on;

title('Magnitude Spectrum of DTFT');

xlabel('\omega (rad/sample)');

ylabel('|X(e^{j\omega})|');
```

Output waveform:



Result:

The Discrete-Time Fourier Transform (DTFT) of the sequence x(n)={1,2,3,4} was computed using MATLAB. The magnitude spectrum was obtained and plotted, illustrating the frequency domain representation of the discrete-time signal.